

Subject Description Form

Subject Code	EIE4114
Subject Title	Digital Forensics for Crime Investigation
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with basic concepts about digital forensic techniques for crime investigation 2. To understand how different forensic techniques are used for information security
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand different approaches for digital forensics 2. Use different techniques for forensic investigation <u>Category B: Attributes for all-roundedness</u> 3. Present ideas and findings effectively
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Digital and Computational Forensics Context</u> Introduction to digital and computational forensics; Historical aspects in digital and computational forensics; Different classes of techniques for forensics: basic idea, framework and applications; Types of forensic evidence with examples 2. <u>The Forensic Process</u> Basic forensic framework; digital evidence identification; triage process, data analysis techniques and forensic report writing 3. <u>Forensics based on Intrinsic/Extrinsic Data</u> Models of digital data capturing device; idea of the use of intrinsic data in digital forensic investigation; introduction to forensics techniques using intrinsic data; applications in source device identification, device linking and integrity verification. Introduction to techniques for multimedia content protection and authentication; attacks modelling. 4. <u>Machine Learning Forensics</u> Different types of ML-based Forensics; Extractive Forensics; Inductive forensics; deductive forensics. Example use cases in ML-based Forensics. 5. <u>Digital Evidence</u> Models of digital evidence; event analytics: surveillance, monitoring, forensic and security; file carving: idea, different classes of techniques; software tools for file carving

	6. <u>Robustness of Forensic Techniques</u> Robustness and security of forensic techniques; adversary model; case studies of reliabilities of forensic techniques. Laboratory Experiments: Practical Works: 1. Study of file carving techniques. 2. Study of source camera identification techniques. 3. Study of Forensic analysis. 4. Study of machine learning and AI forensics analysis																																										
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	The continuous assessment consists of tests, laboratory exercises and a mini-project. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <table><tr><th>Specific Assessment Methods/Tasks</th><th>Remark</th></tr><tr><td>Tests and examination</td><td>end-of chapter type problems used to evaluate students’ ability in applying concepts and skills learnt in the classroom; students need to think critically in order to come with a solution for a problem.</td></tr><tr><td>Laboratory sessions, mini-project</td><td>oral examination will be conducted to evaluate student’s technical knowledge and communication skills.</td></tr></table>		Specific Assessment Methods/Tasks	Remark	Tests and examination	end-of chapter type problems used to evaluate students’ ability in applying concepts and skills learnt in the classroom; students need to think critically in order to come with a solution for a problem.	Laboratory sessions, mini-project	oral examination will be conducted to evaluate student’s technical knowledge and communication skills.
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Student Study Effort Expected	Class contact (time-tabled):							
	• Lecture	21 Hours						
	• Tutorial/Laboratory/Practice Classes	18 Hours						
	Other student study effort:							
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours						
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours						
	Total student study effort:	105 Hours						
Reading List and References	Reference Books: - Joakim Kavrestad, Marcus Birath, and Nathan Clarke, “Fundamentals of Digital Forensics: A Guide to Theory, Research and Applications”, Springer International Publishing AG, third edition, 2024. - Dmitry Vostokov, “Fundamentals of Trace and Log Analysis: a Pattern-oriented Approach to Monitoring, Diagnostics, and Debugging”, Apress, 2023. - Marwan Omar and Hewa Majeed Zangana, “Digital Forensics in the Age of AI”, Hershey PA: IGI Global, 979-833730856, 2025. - Adam Tilmar Jakobsen, “Practical Cyber Intelligence: A Hands-on Guide to Digital Forensics”, Wiley, 2024. - Sabyasachi Pramanik, “Cyber Security and Digital Forensics: Challenges and Future”, John Wiley & Sons, Inc., 2022. - Darren R. Hayes, “<i>A Practical Guide to Digital Forensics Investigations</i>”, Pearson IT Certification, 2020. - Nhien-An Le-Khac, Kim-Kwang Raymond Choo, “Cyber and Digital Forensic Investigations: A Law Enforcement Practitioner’s Perspective”, Springer International Publishing AG, first edition, 2020. - Nihad A Hassan, “<i>Digital Forensics Basics: A Practical Guide using Windows OS</i>”, Apress 2019. - Anders Flaglien, Inger Marie Sunde, AusraDilijonaite, Jeff Hamm, Hens Petter Sandvik, PetterBjelland, Katrin Franke, Stefan Axelsson, “<i>Digital Forensics: an academic introduction</i>”, John Wiley & Sons, 2018.							

	10. Li Chang-Tsun, “<i>Emerging Digital Forensics Applications for Crime Protection, Prevention and Security</i>”, IGI Global 2013, doi:10.4018/978-1-4666-4006-1, 2013. 11. Li Chang-Tsun and Anthony T.S. Ho, “<i>Crime Prevention Technologies and Applications for Advancing Criminal Investigation</i>”, IGI Global 2012, doi:10.4018/978-1-4666-1758-2, 2012
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July 2025